

Laplace transform & Z transform

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SDU Robotics

Topics to be covered in this course

- Sampling and reconstruction
- Aliasing
- Quantization and dynamic range
- Implementation
- Conversion time-frequency domain
- **Z transform**
- Linear Time Invariant system (LTI)
- System analysis
- Window functions
- Filter design
- Impulse response (FIR and IIR)

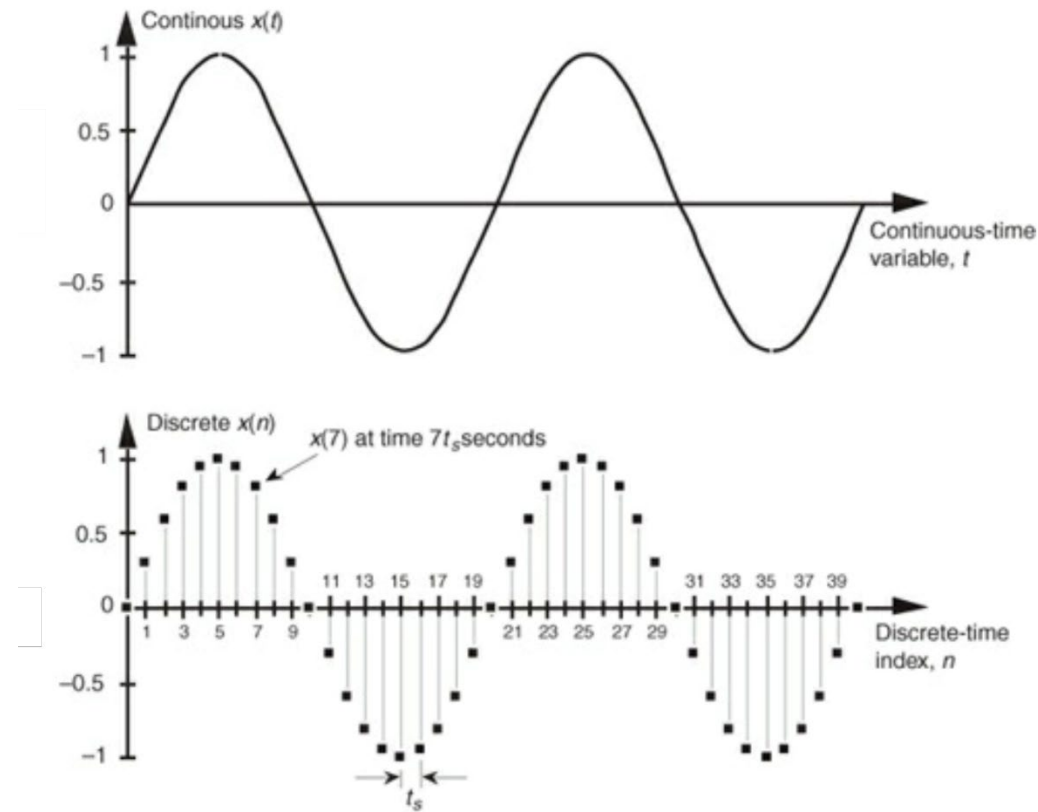
The z-Transform

5

Reference book

Chapter 5: The z-Transform
(Page 137-160)

Discrete time signal v.s. continuous time signal



Discrete time signal v.s. continuous time signal

For discrete time signal $x(n)$,
the **z-transform** of a causal sequence $x(n)$ is defined as

$$X(z) = \mathcal{Z}\{x(n)\} = \sum_{n=0}^{\infty} x(n)z^{-n}$$

For continuous time signal $x(t)$,
Laplace transform of a signal $x(t)$ is defined as

$$X(s) = \mathcal{L}\{x(t)\} = \int_0^{\infty} x(t)e^{-st} dt$$

Discrete time system v.s. continuous time system

Discrete time system

Transfer function of a discrete time system:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{i=0}^N a_i z^{-i}}{1 + \sum_{i=1}^N b_i z^{-i}}$$

Continuous time system

Transfer function of a continuous time system

$$H(s) = \frac{Y(s)}{X(s)} = \frac{\sum_{i=0}^N a_i s^i}{1 + \sum_{i=1}^N b_i s^i}$$

$X()$ represents the input signal,
 $Y()$ represents the output signal

Complex Fourier transform -> Laplace transform

Complex Fourier transform:

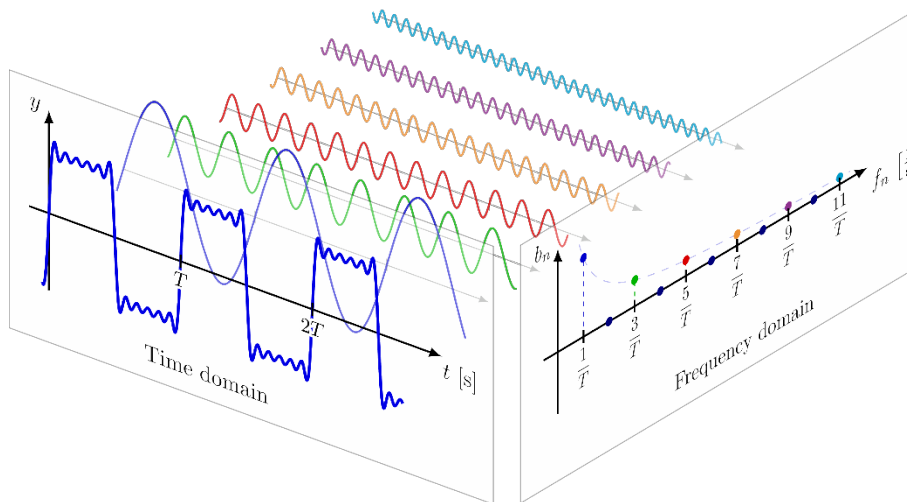
$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

It can be extended to Laplace transform

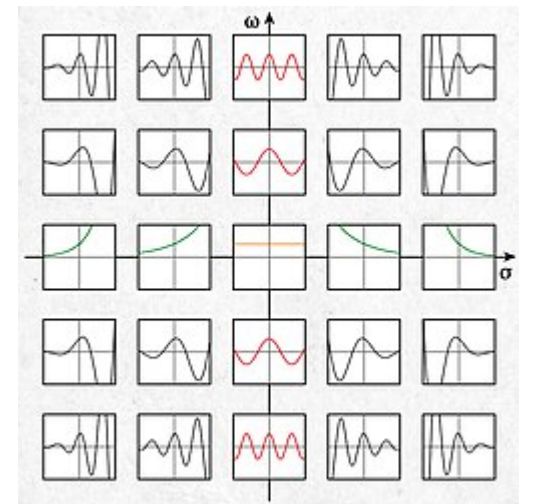
$$X(\sigma, \omega) = \int_{-\infty}^{\infty} [x(t)e^{-\sigma t}]e^{-j\omega t} dt$$

What does $e^{-\sigma t}$ mean?

Fourier
Transform



Laplace
Transform



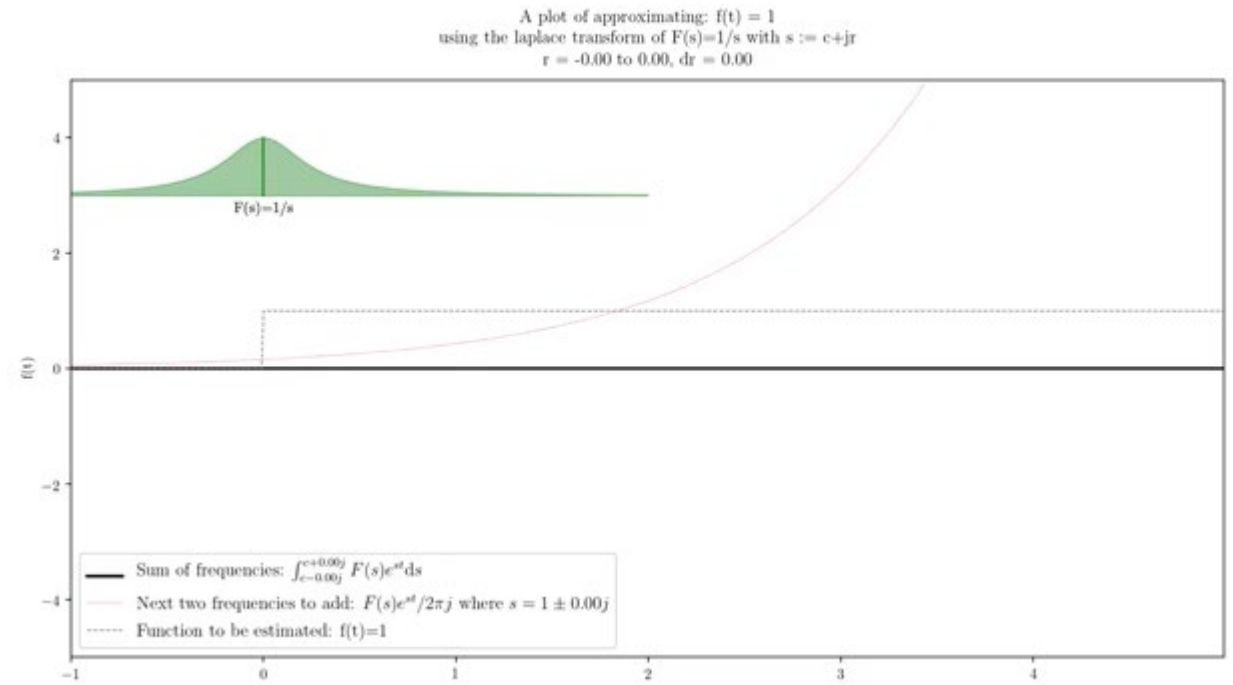
Laplace transform and S domain

The **Laplace transform** changes a signal in the time domain into a signal in the **s-domain**.

$$X(\sigma, \omega) = \int_{-\infty}^{\infty} x(t) e^{-(\sigma + j\omega)t} dt$$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

where $s = \sigma + j\omega$



Example

Let's consider a **continuous** signal

$$x(t) = e^{-at}$$

Its Laplace transform is

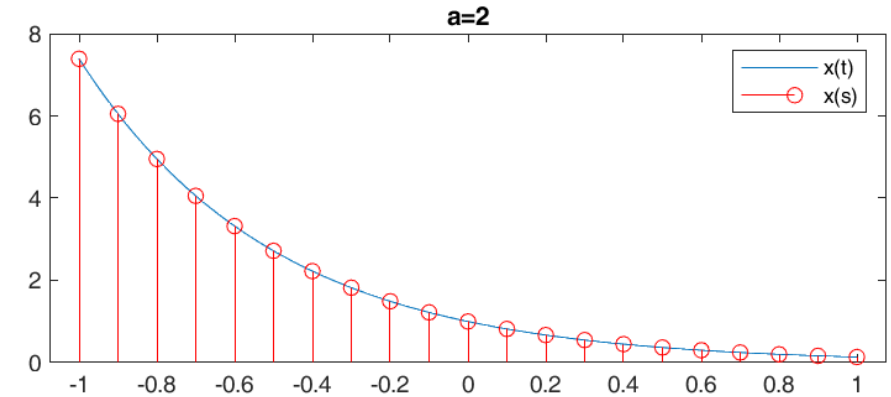
$$X(s) = \frac{1}{s + a}$$

After impulse sampling, we receive the following sequence

$$x(nT) = e^{-anT}, \quad a > 0$$

Laplacian transform

$$\begin{aligned} X_s(s) &= \sum_{n=0}^{\infty} x(nT) e^{-snT} \\ &= \sum_{n=0}^{\infty} e^{-anT} e^{-snT} = \sum_{n=0}^{\infty} e^{-(s+a)nT} \end{aligned}$$



For $|x| < 1$, infinite quotient series

$$\sum_{i=0}^{\infty} x^i = \frac{1}{1 - x}$$

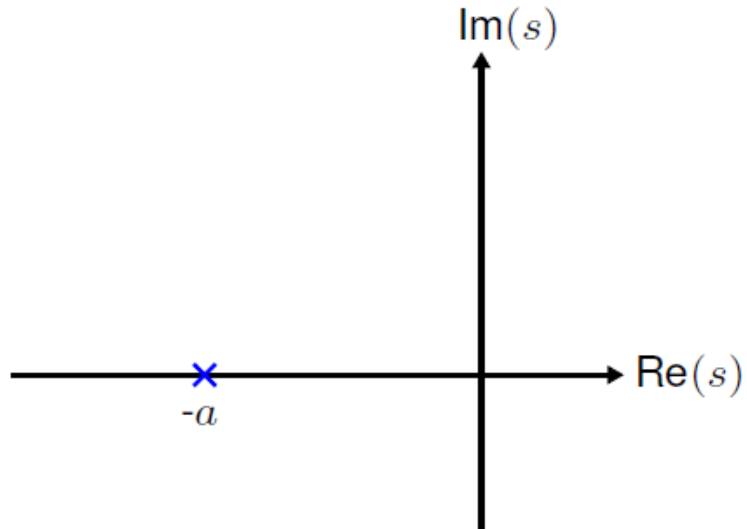
It can be rewritten as

$$X_s(s) = \frac{1}{1 - e^{-(s+a)T}}$$

Continuous signal

$$X(s) = \frac{1}{s + a}$$

This is a transfer function with one **pole** in $s = -a$



Impulse sampled sequence

$$X_s(s) = \frac{1}{1 - e^{-(s+a)T}}$$

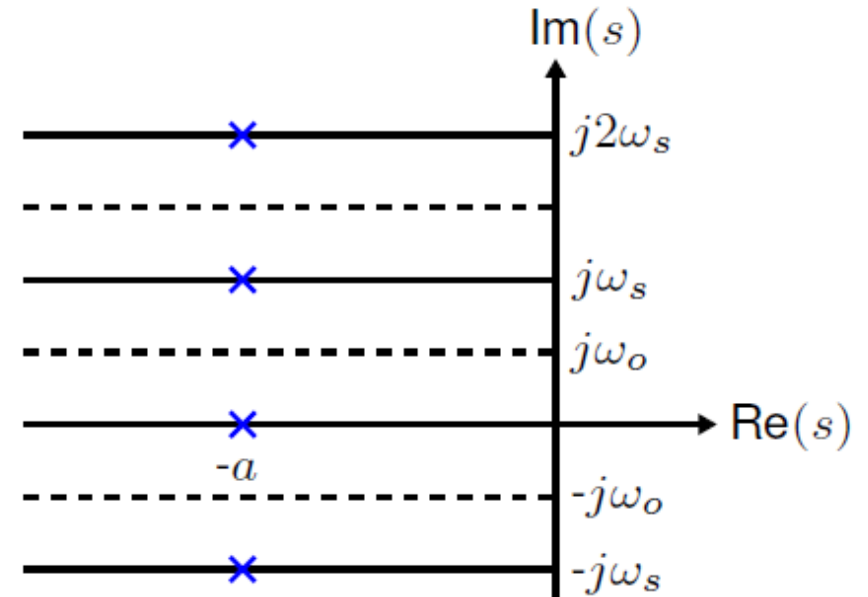
It has **poles**

$$s = -a \pm jm \frac{2\pi}{T} = -a \pm jm 2\pi f_s$$

$$e^{-(s+a)T} = 1$$

Euler's identity

$$e^{2m\pi} = 1$$



When sampling, the pole-zero diagram along the imaginary axis is repeated periodically with the sample frequency.

Relation between s-domain and z-domain

Laplace transform of **sequence** $x(n)$

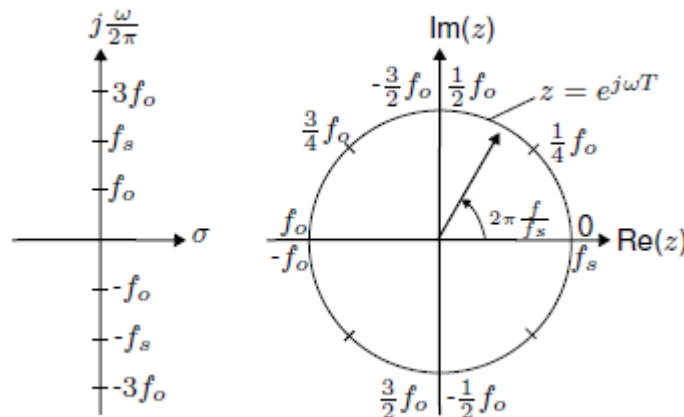
$$X_s(s) = \sum_{n=0}^{\infty} x(n)e^{-snT}$$

z transform of **sequence** $x(n)$

$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$$

$$X_s(s) = X(z) \text{ when } z = e^{sT}$$

Where $s = \sigma + j\omega$



Q: What f_0 here mean?

s domain

z domain

Z transform

For discrete signal process, we use z transform

Definition

The z transform of a sequence $x(n)$ is defined by

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

Notation $X(z) = \mathcal{Z}[x(n)]$

The inverse z transform

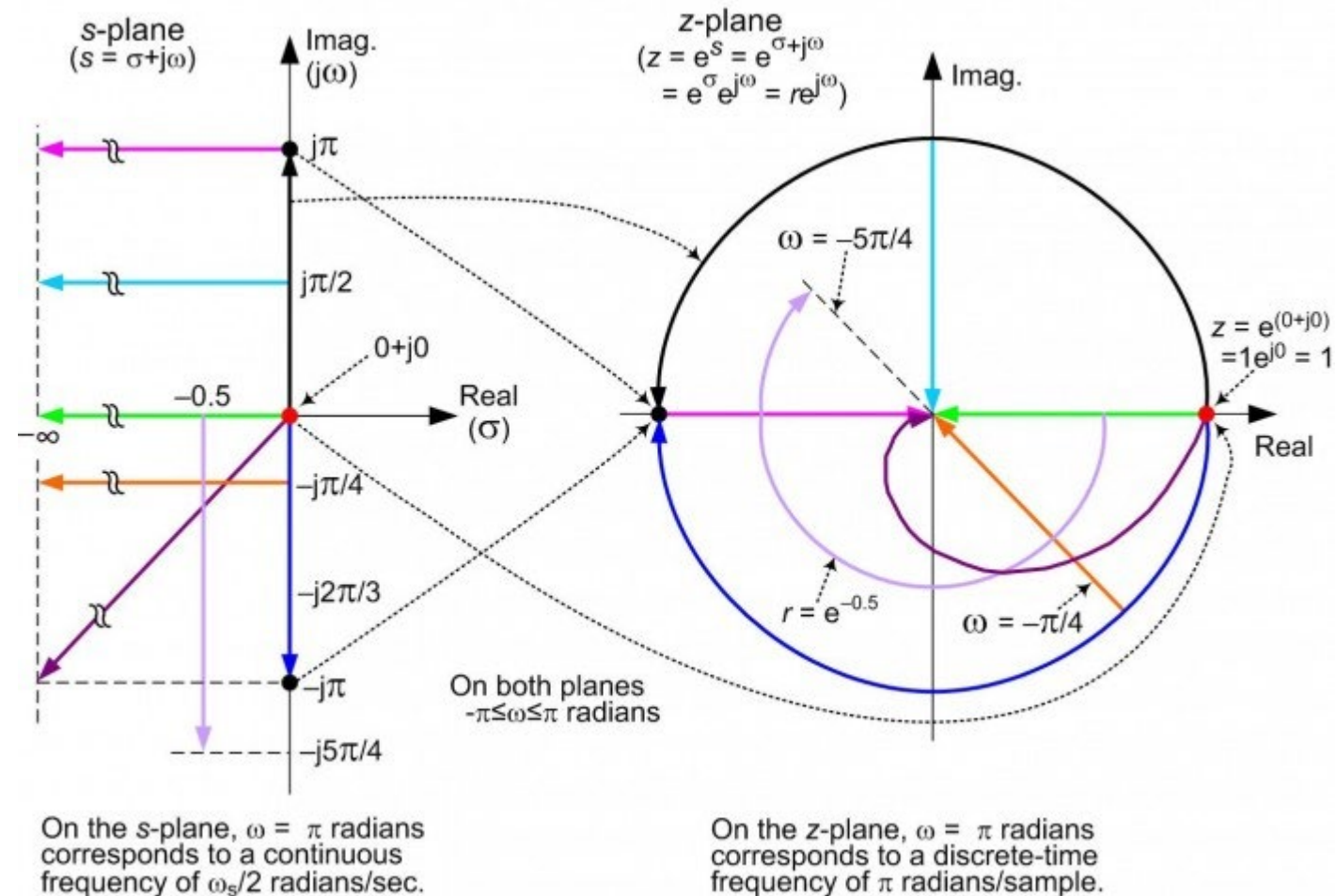
$$x(n) = \mathcal{Z}^{-1}[X(z)]$$

Relation between s plane and z plane

The relation between s plane and z plane as follows

$$z = e^{sT} = e^{\sigma/f_s} \angle 2\pi \frac{f}{f_s}$$

z-plane is a polar coordinate!



Imagine axis of s-plane

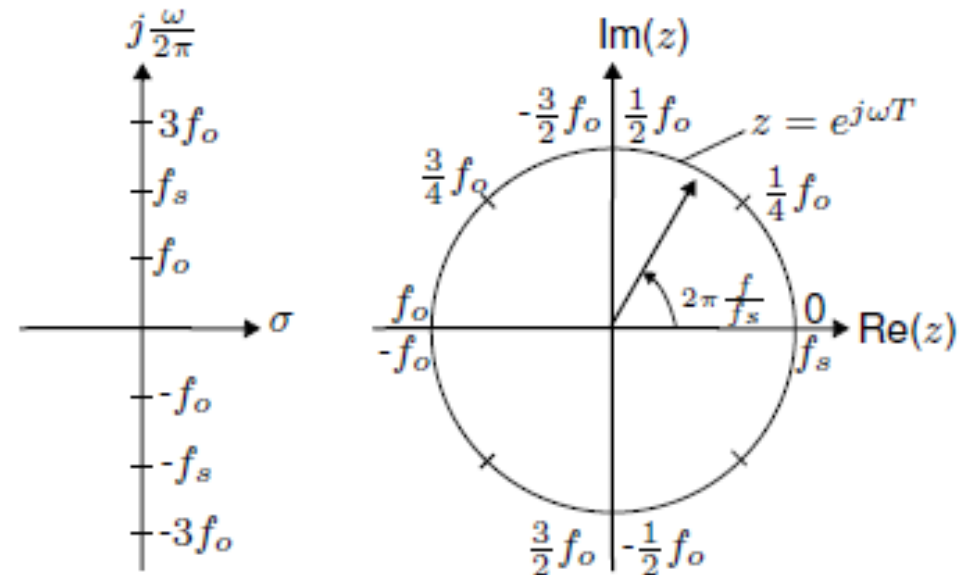
The Imaginary axis of the s-plane can be understood as a set where $\sigma = 0$
In this case,

$$z = e^{j\omega T}$$

When ($\omega = 2\pi f$)

$$z = 1 \angle 2\pi \frac{f}{f_s}$$

$$s = \sigma + j\omega$$



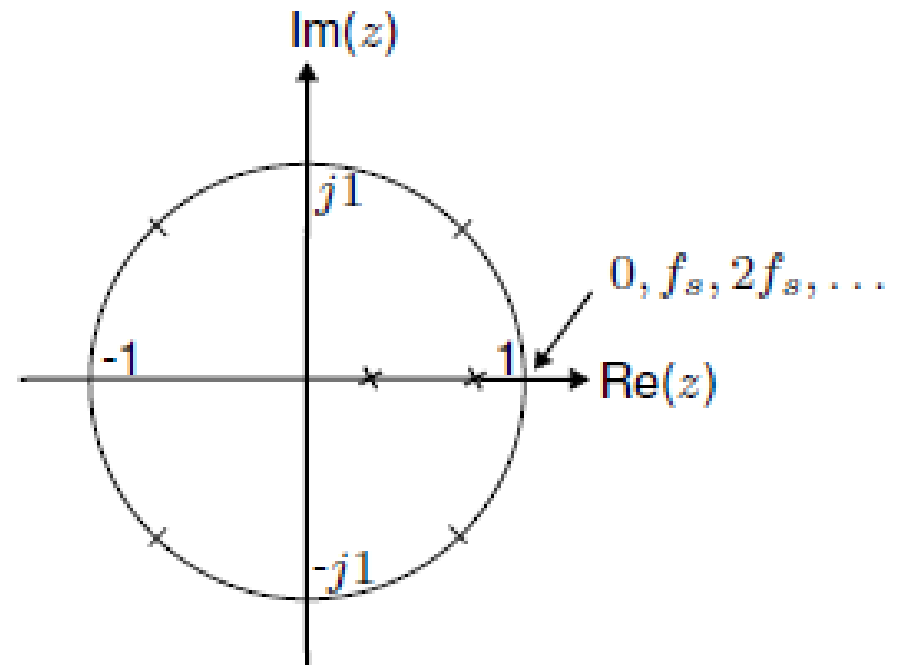
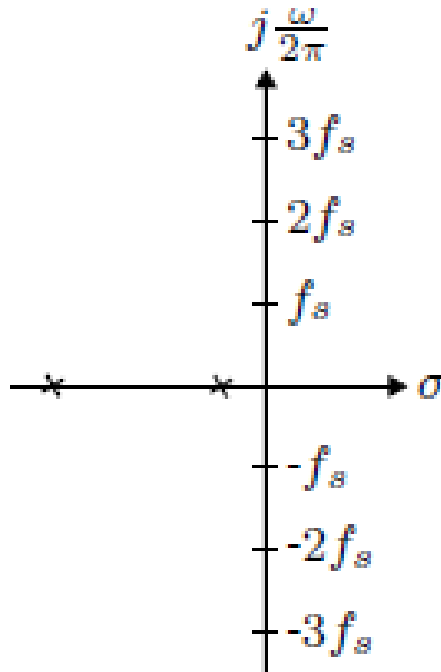
Real axis of s-plane

The real axis of the s-plane can be understood as a set where $\omega = 0$

$$z = e^{\sigma/f_s} \angle 0^\circ$$

$$s = \sigma + j\omega$$

Q: only the positive Re(z)? Or the whole axis of Re(z)?

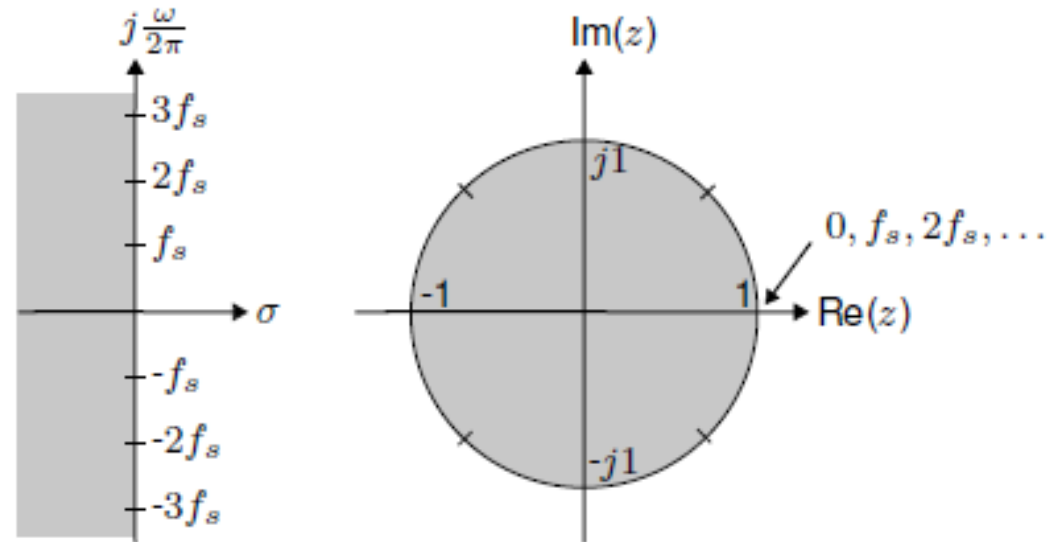


Left half-plane of s-plane

The left half-plane of s-plane $\rightarrow \sigma < 0$

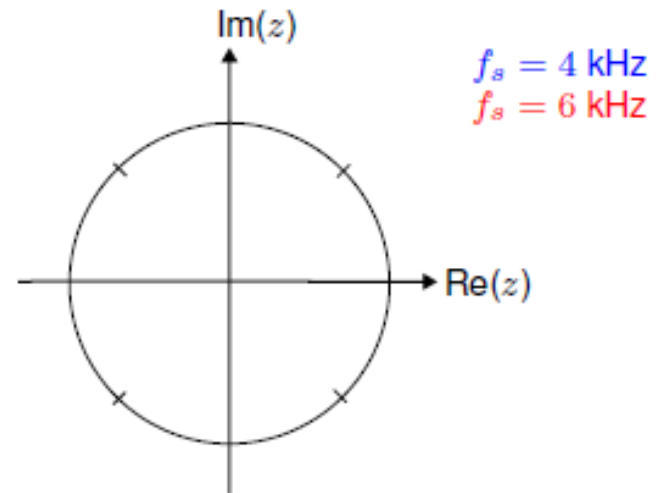
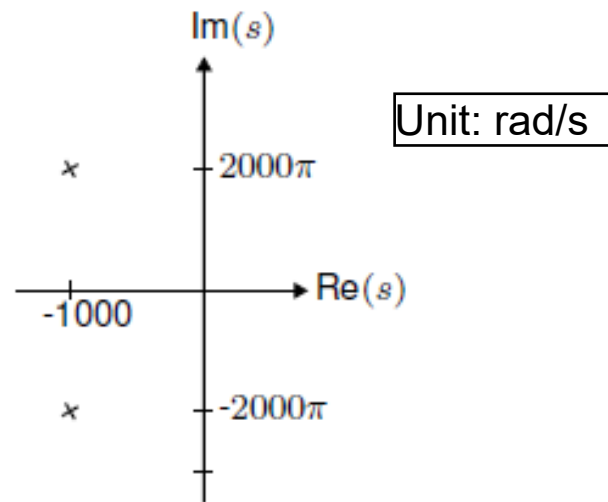
It means that $|z| = e^{\sigma/f_s} < 1$.

Thus, the left half-plane becomes the interior of the unit circle.

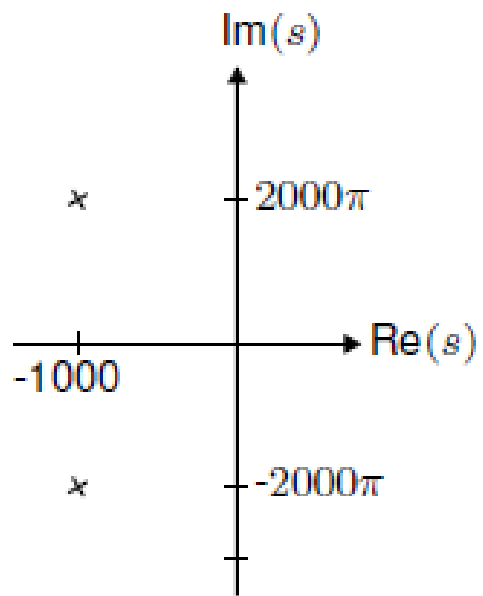


Exercise

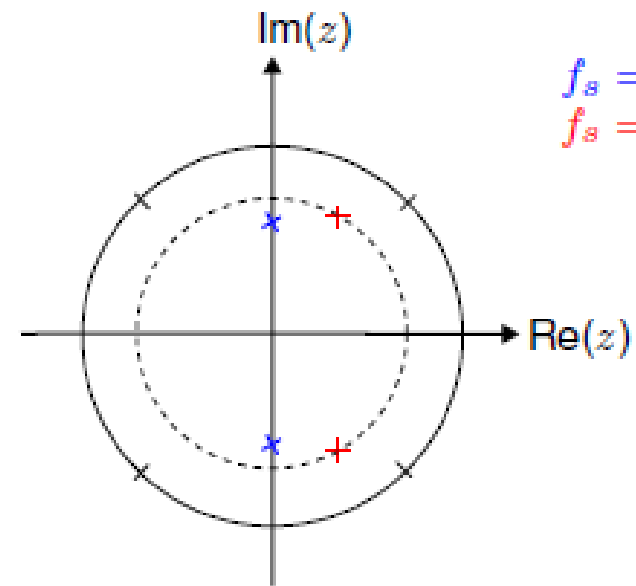
Where in the z plane are the poles when $f_s = 4$ kHz and $f_s = 6$ kHz?



$$z = e^{sT} = e^{\sigma/f_s} \angle 2\pi \frac{f}{f_s}$$



Unit: rad/s



$f_s = 4 \text{ kHz}$
 $f_s = 6 \text{ kHz}$

Converge

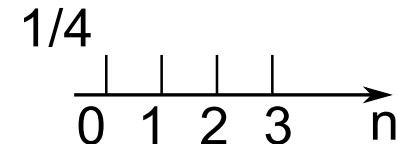
Region of convergence (ROC) = $\{z | \sum_n x(n)z^{-n} \text{ converge}\}$

Where z is complex number.

Example 1

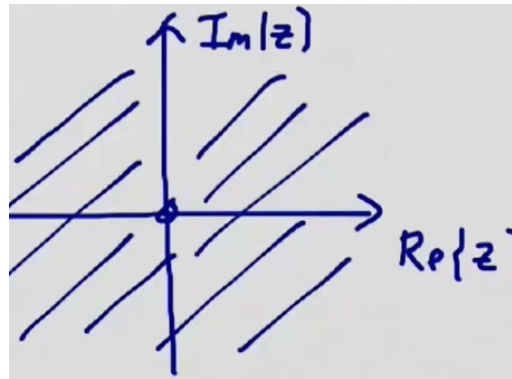
4 points rectangular window: $x(n) = \frac{1}{4} \sum_{k=0}^3 \delta(n - k)$

$$X(z) = \frac{1}{4} (1 + z^{-1} + z^{-2} + z^{-3})$$

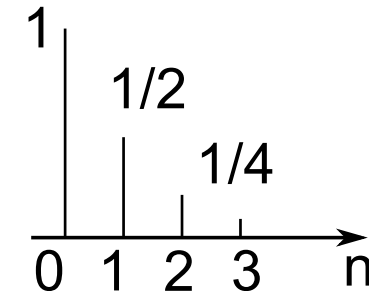


ROC: the range of z that makes $X(z)$ converge?

$$ROC = \{z | z \neq 0\}$$



Example 2

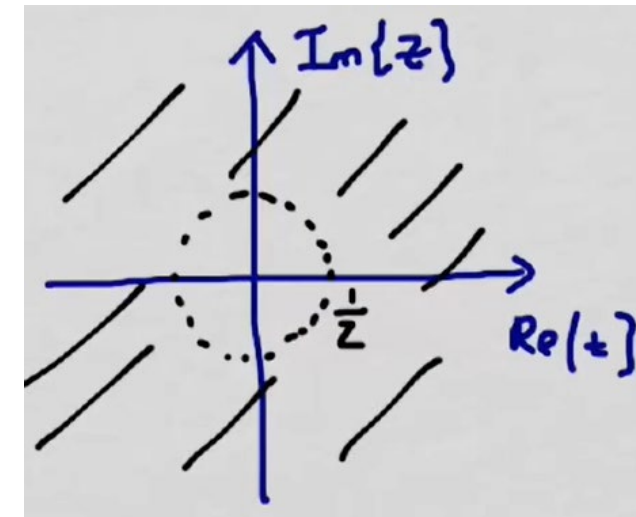


$$x(n) = \left(\frac{1}{2}\right)^n$$

$$X(Z) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} = \sum_{n=0}^{\infty} \left(\frac{1}{2z}\right)^n$$

$X(Z)$ converge if $\left|\frac{1}{2z}\right| < 1$,

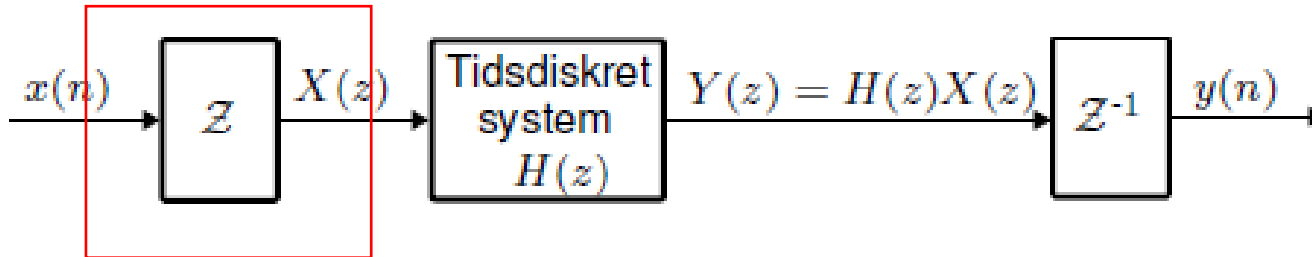
$$ROC = \left\{z \mid |z| > \frac{1}{2}\right\}$$



Z domain in discrete signal processing

Process

- Perform z transform on input signal $x(n)$
- Describe the discrete time system in z domain $H(z)$
- Obtain output $Y(z) = H(z)X(z)$
- Perform inverse z transform on $Y(z)$ to obtain $y(n)$



Z-transform rules

A number of transform rules can facilitate the calculation of the z-transform
Especially the rules Z1 and Z2 are often used in this course.

Regel	$x(n)$	$X(z)$
Z1	$ax_1(n) + bx_2(n)$	$aX_1(z) + bX_2(z)$
Z2	$x(n - m)$	$z^{-m}X(z)$
Z3	$x(n)a^{-n}$	$X(az)$
Z4	$x(n)s^{-bn}$	$X(e^{bT}z)$
Z5	$\sum_{m=0}^n x(m)h(n - m)$	$X(z)H(z)$

Prove

The z-transform of $x(n - m)$ is

$$\mathcal{Z}[x(n - m)] = \sum_{n=0}^{\infty} x(n - m)z^{-n}$$

where $x(n)$ is a **causal sequence**.

Therefore, the above can be written

$$\mathcal{Z}[x(n - m)] = \sum_{n=m}^{\infty} x(n - m)z^{-n}$$

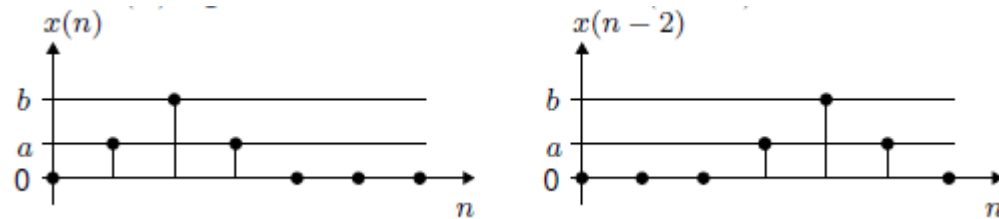
By defining $k := n - m$, we have

$$\mathcal{Z}[x(n - m)] = \sum_{k=0}^{\infty} x(k)z^{-(k+m)} = z^{-m} \sum_{k=0}^{\infty} x(k)z^{-k}$$

$$\mathcal{Z}[x(n - m)] = z^{-m}X(z)$$

Example

Let's compare the sequence $x(n)$ and the delayed sequence $x(n - 2)$ in the z-domain.



You can determine the z-transform of $x(n]$ as

$$X(z) = \mathcal{Z}[x(n)] = \sum_{n=0}^{\infty} x(n)z^{-n}$$

From the graph it can be seen that

$$X(z) = a z^{-1} + b z^{-2} + a z^{-3}$$

$$\mathcal{Z}[x(n-2)] = z^{-2} \sum_{n=0}^{\infty} x(n)z^{-n} = a z^{-3} + b z^{-4} + a z^{-5}$$

Common z-transform for signal

Par	$x(n)$	$X(z)$
ZT1	$\delta(n)$	1
ZT2	$u(n)$	$\frac{z}{z-1}$
ZT3	n	$\frac{z}{(z-1)^2}$
ZT4	a^n	$\frac{z}{z-a}$
ZT5	$e^{s_0 n T}$	$\frac{z}{z-e^{s_0 T}}$
ZT6	$\sin \omega_0 n T$	$\frac{(\sin \omega_0 T)z}{z^2 - 2(\cos \omega_0 T)z + 1}$
ZT7	$\cos \omega_0 n T$	$\frac{z^2 - (\cos \omega_0 T)z}{z^2 - 2(\cos \omega_0 T)z + 1}$

Example

Calculate the z-transform of **unit sequence** $u(n)$

$$U(z) = Z[u(n)] = \sum_{n=0}^{\infty} u(n)z^{-n}$$

Since $u(n) = 1$ is for $n \geq 0$, so we have

$$U(z) = \sum_{n=0}^{\infty} z^{-n}$$

Therefore,

$$U(z) = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1}$$

Geometric series sum:

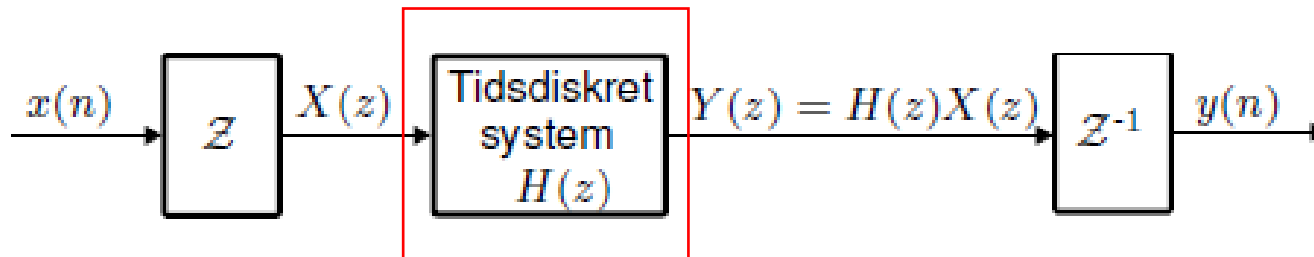
$$\sum_{i=0}^{\infty} x^i = \frac{1}{1 - x}$$

Transfer function in z domain

Discrete systems can be described by a transfer function as

$$H(z) = \frac{Y(z)}{X(z)}$$

$H(z)$ is a transfer function, and $X(z)$, $Y(z)$ is the input and output sequence



Recap:

$$\begin{aligned}y(n) &= ay(n-1) + x(n) \\x(n) &= \delta(n)\end{aligned}$$

In this case, $y(0) = ay(-1) + x(0)$

Then

$$y(-1) = 0, \quad y(n) = 0, \quad n < 0$$

$$y(0) = 1$$

And

$$\begin{aligned}y(1) &= ay(0) + x(1) = a \\y(2) &= ay(1) + x(2) = a^2 \\&\dots\end{aligned}$$

Results

$$y(n) = \begin{cases} 0, & n < 0 \\ a^n, & n \geq 0 \end{cases}$$

Discrete time system: Transfer function

A N th order difference equation describing a causal system can be written as

$$y(n) + b_1y(n-1) + b_2y(n-2) + \dots + b_Ny(n-N) = a_0x(n) + a_1x(n-1) + \dots + a_Nx(n-N)$$

where $x(n-i)$ is the input sequence, $y(n-i)$ is the output sequence,
 a_i, b_i are real coefficients.

The above difference equation can be written as

$$y(n) = \sum_{i=0}^N a_i x(n-i) - \sum_{i=1}^N b_i y(n-i)$$

If a b -coefficient is different from zero, then the differential equation is called a recursive algorithm.

We will talk more about this in FIR and IIR

Transfer function in z domain

A transfer function is found by z-transform of a differential equation of the form

$$y(n) + \sum_{i=1}^N b_i y(n-i) = \sum_{i=0}^N a_i x(n-i)$$

Recall: rule is used for the z-transform

$$z^{-m}X(z) = \mathcal{Z}(x(n-m))$$

Therefore, we have

$$Y(z) \left(1 + \sum_{i=1}^N b_i z^{-i} \right) = X(z) \sum_{i=0}^N a_i z^{-i}$$

The transfer function thus becomes

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{i=0}^N a_i z^{-i}}{1 + \sum_{i=1}^N b_i z^{-i}}$$

It can be rewritten as

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{i=0}^N a_i z^{N-i}}{z^N + \sum_{i=1}^N b_i z^{N-i}}$$

Example: 1st and 2nd order transfer function

A first order system with difference equation

$$y(n) = a_0x(n) + a_1x(n-1) - b_1y(n-1)$$

Its transfer function after z-transform

$$Y(z)(1 + b_1z^{-1}) = X(z)(a_0 + a_1z^{-1})$$
$$H(z) = \frac{Y(z)}{X(z)} = \frac{a_0 + a_1z^{-1}}{1 + b_1z^{-1}} = \frac{a_0z + a_1}{z + b_1}$$

A second order system with difference equation

$$y(n) = a_0x(n) + a_1x(n-1) + a_2x(n-2) - b_1y(n-1) - b_2y(n-2)$$

Its transfer function after z-transform

$$Y(z)(1 + b_1z^{-1} + b_2z^{-2}) = X(z)(a_0 + a_1z^{-1} + a_2z^{-2})$$
$$H(z) = \frac{Y(z)}{X(z)} = \frac{a_0 + a_1z^{-1} + a_2z^{-2}}{1 + b_1z^{-1} + b_2z^{-2}} = \frac{a_0z^2 + a_1z + a_2}{z^2 + b_1z + b_2}$$

Poles and zeros of a transfer function

A function is called a transfer function if it can be written in the form as

$$H(z) = \frac{P(z)}{Q(z)}$$

Both $P(z)$ and $Q(z)$ are polynomials in z

The roots of $P(z)$ are called **zeros** of $X(z)$

The roots of $Q(z)$ are called **poles** of $X(z)$

Example

Considering the following

$$H(z) = \frac{0.58 - 0.58z^{-1}}{1 - 0.16z^{-1}}$$

The transfer function can be rewritten as

$$H(z) = 0.58 \frac{z - 1}{z - 0.16}$$

zero for $H(z)$ is $z = 1$,

pole for $H(z)$ is $z = 0.16$

Pole-Zero diagram

Let's consider the following transfer function

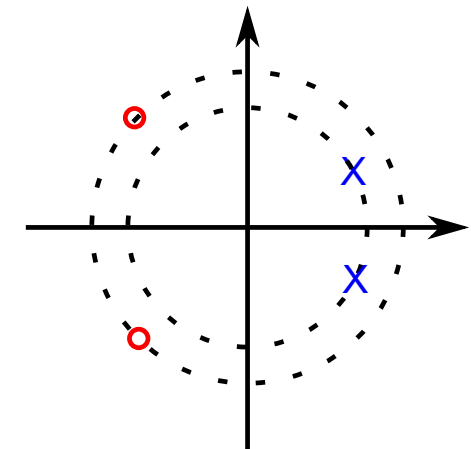
$$H(z) = \frac{0.5 + 0.707z^{-1} + 0.5z^{-2}}{1 - 1.386z^{-1} + 0.64z^{-2}} = 0.5 \frac{z^2 + 1.414z + 1}{z^2 - 1.386z + 0.64}$$

The zeros are found from the roots of the numerator polynomial

$$z^2 + 1.414z + 1 = 0 \quad \Rightarrow \quad z = -0.707 \pm j0.707$$

The poles are found from the roots of the denominator polynomial

$$z^2 - 1.386z + 0.64 = 0 \quad \Rightarrow \quad z = 0.693 \pm j0.4$$



Factorization

From the fundamental theorem of algebra, a transfer function can be written as

$$H(z) = a_0 \frac{(z - z_1)(z - z_2) \dots (z - z_N)}{(z - p_1)(z - p_2) \dots (z - p_N)}$$

Where z_i for $i = 1, \dots, N$ are the zeros of the transfer function

p_i for $i = 1, \dots, N$ are the poles of the transfer function

Matlab function – roots()

returns the roots of the polynomial

For example, $3x^2 - 2x - 4 = 0$

```
p = [3, -2, -4];
```

```
r = roots(p)
```

```
r = 2x1  
  
    1.5352    x1  
   -0.8685    x2
```

Matlab function: tf2zp()

Find the zeros, poles, and gain of the system.

Let's consider the transfer function in the previous example:

$$H(z) = \frac{0.5 + 0.707z^{-1} + 0.5z^{-2}}{1 - 1.386z^{-1} + 0.64z^{-2}} = 0.5 \frac{z^2 + 1.414z + 1}{z^2 - 1.386z + 0.64}$$

Z should be in positive power!!

```
b=[0.5, 0.5*1.414, 0.5*1]; % numerators  
a=[1, -1.386, 0.64]; % denominators  
[z,p,k] = tf2zp(b,a);
```

$$H(z) = 0.5 \frac{(z - (-0.707 + 0.7072i))(z - (-0.707 - 0.7072i))}{(z - (0.693 + 0.3997i))(z - (0.693 - 0.3997i))}$$

```
>> z  
  
z =  
  
-0.7070 + 0.7072i  
-0.7070 - 0.7072i  
  
>> p  
  
p =  
  
0.6930 + 0.3997i  
0.6930 - 0.3997i  
  
>> k  
  
k =  
  
0.5000
```

Matlab – zplane()

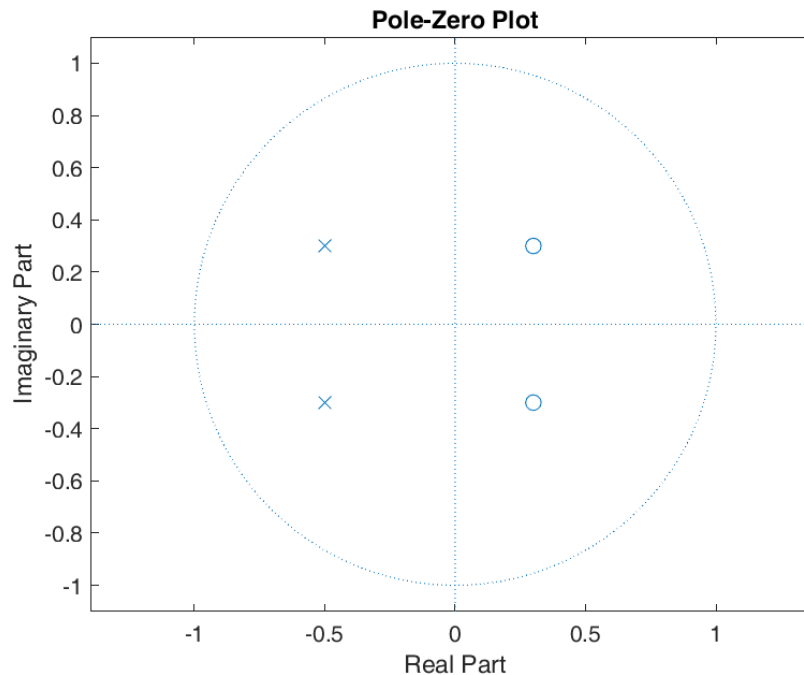
Zero-pole plot for discrete-time systems

zplane(z, p) – plots the zeros and poles defined by z and p, which should be as **column vectors**.

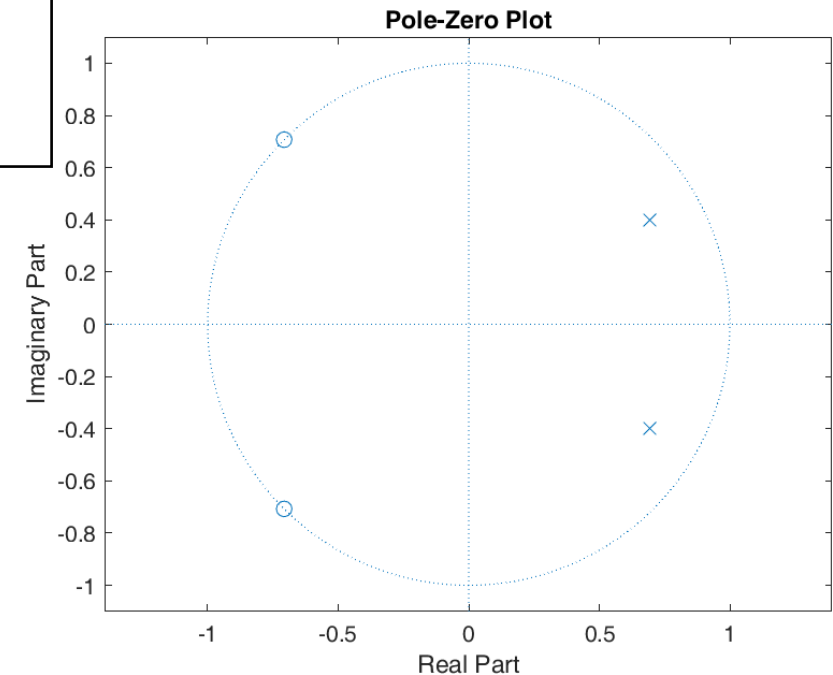
zplane(b, a) – plots the zeros and poles defined by a transfer function whose numerator and denominator are b and a.

Note that b and a should be as **row vectors**.

```
z=[0.3+0.3i; 0.3-0.3i];  
p=[-0.5+0.3i; -0.5-0.3i];  
zplane(z,p);
```



```
b=[0.5, 0.5*1.414, 0.5*1];  
a=[1, -1.386, 0.64];  
zplane(b,a)  
% it is the previous  
example
```

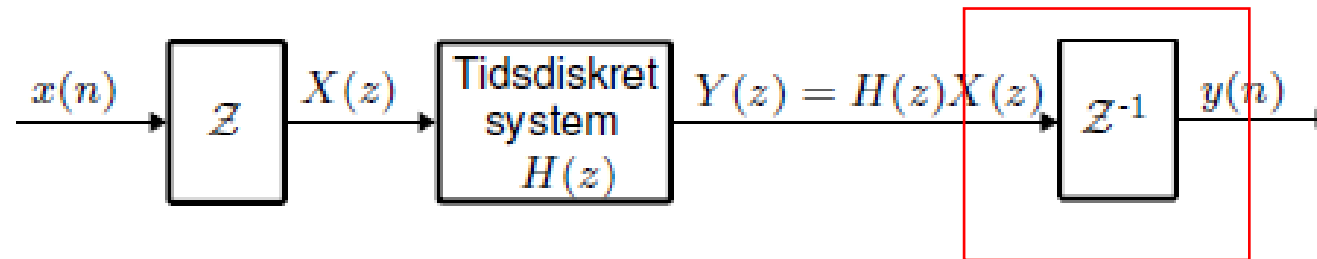


Inverse z-transform

Inverse z-transformation is used to determine the output response $y(n)$ of a discrete-time system for a given input $x(n)$.

The processing of discrete system (**RECAP**):

1. The input sequence $x(n)$ is z-transformed.
2. The system's transfer function $H(z)$ is set up with positive powers of z
3. The output response in z-domain is calculated $Y(z) = H(z)X(z)$
4. The output sequence $y(n)$ is calculated by inverse z-transform of $Y(z)$.



Example

Considering the following transfer function

$$H(z) = \frac{1}{1 - 0.5z^{-1}}$$

We use the inverse z-transform to determine the output response $y(n)$ when the **input $x(n)$ is impulse $\delta(n)$** .

1. The input sequence $x(n) = \delta(n)$ is z-transformed using lookup table

$$X(z) = \mathcal{Z}[x(n)] = \mathcal{Z}[\delta(n)] = 1$$

2. The system's transfer function $H(z)$ is set up with positive powers of z

$$H(z) = \frac{z}{z - 0.5}$$

3. The output response $Y(z)$ is calculated

$$Y(z) = H(z)X(z) = \frac{z}{z - 0.5} \cdot 1$$

4. According to the lookup table (ZT4)

$$y(n) = \mathcal{Z}^{-1}\left[\frac{z}{z - 0.5}\right] = 0.5^n$$

Par	$x(n)$	$X(z)$
ZT1	$\delta(n)$	1
ZT2	$u(n)$	$\frac{z}{z-1}$
ZT3	n	$\frac{z}{(z-1)^2}$
ZT4	a^n	$\frac{z}{z-a}$
ZT5	$e^{s_0 nT}$	$\frac{z}{z-e^{s_0 T}}$
ZT6	$\sin \omega_0 nT$	$\frac{(\sin \omega_0 T)z}{z^2-2(\cos \omega_0 T)z+1}$
ZT7	$\cos \omega_0 nT$	$\frac{z^2-(\cos \omega_0 T)z}{z^2-2(\cos \omega_0 T)z+1}$

Partial fraction

The procedure for partial fraction solution is

1. Set up expressions for $Y(z)$ with positive power of z in factorization form

$$Y(z) = \frac{T(z)}{N(z)} = \frac{T(z)}{(z - p_1)(z - p_2) \dots (z - p_N)}$$

2. Divide $Y(z)$ by z so that the order number of the denominator is greater than the order number of the numerator. This expression resolves into fractions

$$\frac{\textcolor{red}{Y}(z)}{\textcolor{red}{z}} = \frac{T(z)}{\textcolor{red}{z}N(z)} = \frac{k_1}{z - p_1} + \frac{k_2}{z - p_2} + \dots + \frac{k_N}{z - p_N}$$

3. Numerator coefficients k_i are calculated as

$$k_i = (z - p_i) \frac{Y(z)}{z} \Big|_{z=p_i}$$

4. Write down $\frac{Y(z)}{z}$ in fractional form and multiply by z .

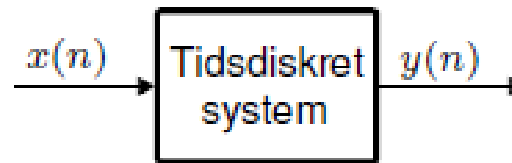
5. Inverse z-transform all the fractions. (from Table)

Par	$x(n)$	$X(z)$
ZT1	$\delta(n)$	1
ZT2	$u(n)$	$\frac{z}{z-1}$
ZT3	n	$\frac{z}{(z-1)^2}$
ZT4	a^n	$\frac{z}{z-a}$
ZT5	$e^{s_0 n T}$	$\frac{z}{z-e^{s_0 T}}$
ZT6	$\sin \omega_0 n T$	$\frac{(\sin \omega_0 T) z}{z^2 - 2(\cos \omega_0 T) z + 1}$
ZT7	$\cos \omega_0 n T$	$\frac{z^2 - (\cos \omega_0 T) z}{z^2 - 2(\cos \omega_0 T) z + 1}$

Example

Let's consider the following differential equation (transfer function):

$$y(n) = x(n) + 0.5y(n - 1)$$



What will be the output sequence $y(n)$ when the input sequence $x(n)$ is a unit sequence $u(n)$ and $y(n) = 0$ for $n < 0$?

We can calculate by hand when $n = 0, 1, 2, \dots$

$$y(0) = x(0) + 0,5y(-1) = 1$$

$$y(1) = x(1) + 0,5y(0) = 1,5$$

$$y(2) = x(2) + 0,5y(1) = 1,75$$

$$y(3) = x(3) + 0,5y(2) = 1,875$$

Example continue...

Considering transfer function

$$H(z) = \frac{z}{z - 0.5}$$

We use inverse z-transformation to determine the output response $y(n)$ when the input stimulus $x(n)$ is an **unit sequence** $u(n)$, i.e.

$$Y(z) = H(z)X(z) = \frac{z}{z - 0.5} \frac{z}{z - 1} = \frac{z^2}{(z - 0.5)(z - 1)}$$

We follow the procedure for partial fraction solution

1. Set up expressions for $Y(z)$ with positive powers of z in factored form

$$Y(z) = \frac{T(z)}{N(z)} = \frac{z^2}{(z - 0.5)(z - 1)}$$

where $z = 0.5$ and $z = 1$ are roots of the denominator polynomial of $Y(z)$

Example continue...

2. Divide $Y(z)$ by z so that the order of the denominator is greater than the order of the numerator

$$\frac{Y(z)}{z} = \frac{z}{(z - 0.5)(z - 1)} = \frac{k_1}{z - 0.5} + \frac{k_2}{z - 1}$$

3. The numerator coefficients are calculated as

$$k_1 = (z - p_1) \frac{Y(z)}{z} \Big|_{z=p_1} = \frac{z}{z - 1} \Big|_{z=0.5} = -1$$

$$k_2 = (z - p_2) \frac{Y(z)}{z} \Big|_{z=p_2} = \frac{z}{z - 0.5} \Big|_{z=1} = 2$$

4. Write $Y(z)$ in fractional form and multiply by z

$$Y(z) = -\frac{1}{z - 0.5} z + \frac{2}{z - 1} z$$

5. Inverse z-transform all the fractions. (Table ZT4)

$$y(n) = \mathcal{Z}^{-1}[Y(z)] = -\mathcal{Z}^{-1}\left[\frac{z}{z - 0.5}\right] + 2\mathcal{Z}^{-1}\left[\frac{z}{z - 1}\right] = -0.5^n + 2 \cdot 1^n = 2 - 0.5^n$$

$$\begin{aligned} y(0) &= 1 \\ y(1) &= 1.5 \\ y(2) &= 1.75 \\ y(3) &= 1.875 \end{aligned}$$

...

$$y(100) = 1.999$$

Compare with the
results 2 slides before!

Matlab: inverse z-transform

Matlab function:

`[r,p,k] = residuez(numerators, denominators)`

Digital system:

$$\frac{B(z)}{A(z)} = \frac{r(1)}{1 - p(1)z^{-1}} + \dots + \frac{r(n)}{1 - p(n)z^{-1}} + k(1) + k(2)z^{-1} + \dots + k(m - n + 1)z^{-(m-n)}$$

`[r,p,k] = residue(numerators, denominators)`

Analog system:

$$\frac{b(s)}{a(s)} = \frac{b_ms^m + b_{m-1}s^{m-1} + \dots + b_1s + b_0}{a_ns^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0} = \frac{r_n}{s - p_n} + \dots + \frac{r_2}{s - p_2} + \frac{r_1}{s - p_1} + k(s).$$

Matlab

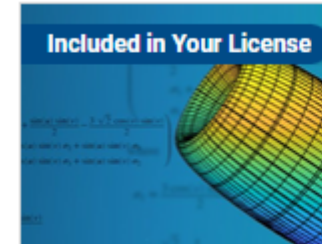
Check functions:

ztrans()

iztrans()

```
syms n
f = sin(n);
fz = ztrans(f)
```

```
syms z
F = 2*z/(z-2)^2;
iztrans(F)
```



Symbolic Math Toolbox

by MathWorks

Perform symbolic math computations

Exercise

5.3. Find the z-transform for each of the following sequences:

a. $x(n) = 3u(n - 4)$

b. $x(n) = 2(-0.2)^n u(n)$

c. $x(n) = 5e^{-2(n-3)} u(n - 3)$

d. $x(n) = 6(0.6)^n \cos(0.2\pi n) u(n)$

e. $x(n) = 4e^{-3(n-1)} \sin(0.2\pi(n - 1)) u(n - 1).$

Exercise

5.5. Given two sequences

$$x_1(n) = 5\delta(n) - 2\delta(n - 2) \text{ and } x_2(n) = 3\delta(n - 3)$$

- a. determine the z-transform of the convolution of the two sequences using the convolution property of z-transform

$$X(z) = X_1(z)X_2(z)$$

- b. determine the convolution by the inverse z-transform

$$x(n) = Z^{-1}(X_1(z)X_2(z))$$

from the result in (a).

Exercise

5.9. Using the partial fraction expansion method, find the inverse of the following z-transforms:

a. $X(z) = \frac{1}{z^2 - 0.3z - 0.24}$

b. $X(z) = \frac{z}{(z - 0.2)(z + 0.4)}$

c. $X(z) = \frac{z}{(z + 0.2)(z^2 - z + 0.5)}$

d. $X(z) = \frac{z(z + 0.5)}{(z - 0.1)^2(z - 0.6)}$